Electromagnetic induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why changing magnetic flux can induce an emf.

SUCCESS CRITERIA - I can...

1. Identify a change in magnetic flux.
2. Predict whether a stationary arrangement produces sustained induction.
3. Use Lenz's law to describe opposition to change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "induction" and "flux" to explain your answer. Sentence starters are optional.

induction: producing an emf through a changing magnetic flux

flux: a measure of magnetic field passing through a chosen area

emf: energy supplied per charge, measured in volts

01 Warm-up: bring back an earlier idea

A **3 Ω** resistor carries **5 A**. **Find power** using both IV and I^2R, showing the implied voltage.

Word help: voltage = electric potential difference; energy transferred per charge

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

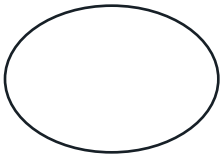
Reference: $\Phi = BA$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$.
Uniform perpendicular field for $\Phi = BA$. N is coil turns; the induced effect opposes the flux change.

Worked reasoning: A changing magnetic flux through a coil can induce voltage. Holding a magnet stationary relative to a fixed coil gives no induction from relative motion in this simple setup.

Your turn: Draw a magnet approaching a coil and **mark the changing field through** the loop. Contrast it with a stationary case.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

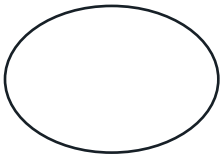
Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Compare a magnet held still** beside a fixed coil with a magnet moving into the coil.
2. Which can produce a changing flux?

Sentence starter: Both ... . They differ in ... because ... .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

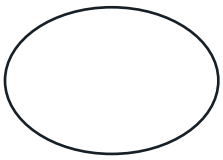
04 Apply the physics

A **100**-turn lab coil has area **0.020 m²**. A uniform field perpendicular to it changes from **0.10 T** to **0.30 T** in **0.50 s**. **Label the magnetic-flux diagram** and **calculate average induced-emf magnitude**.

Word help: uniform = constant throughout the stated interval

Sentence starter: I found ... [unit]. This value means ... in this situation.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why does Lenz’s law oppose the change in flux rather than always oppose the field itself?

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Both charge magnitudes become **4** times as large and separation becomes **6** times as large. Express new force as a fraction of the old.
Label charges and both interaction forces.

Word help: separation = distance between the specified objects or centers

Sentence starter: In my work, I used … to show … . I labeled it … .



Charge A Charge B

Marzano check - circle: 1 2 3

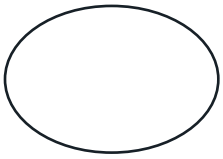
07 Exit check

The same flux change occurs in one fourth the time. **Compare emf magnitude** and **name one thing the calculation** treats as true.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both … . They differ in … because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

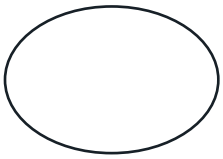
08 STRETCH: test the prediction

Flux per turn changes from **0.01** to **0.04 Wb** in **0.2 s** through **50 turns**. **Find average emf magnitude** and **describe Lenz's-law direction qualitatively**.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3